

Project 6:Wine Quality Prediction

Importing the Dependencies

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

Data Collection

```
In [2]: #Loading the dataset to a pandas DataFrame
wine_dataset=pd.read_csv(r'D:\Projects\archive (5)\wineQT.csv')
```

```
In [3]: #number of rows & columns in the dataset
wine_dataset.shape
```

Out[3]: (1143, 13)

```
In [4]: #first 5 rows of the dataset
wine_dataset.head()
```

Out[4]:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphate
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56

```
In [5]: #Checking for missing values
wine_dataset.isnull().sum()
```

```
Out[5]: fixed acidity      0
        volatile acidity  0
        citric acid       0
        residual sugar    0
        chlorides         0
        free sulfur dioxide 0
        total sulfur dioxide 0
        density           0
        pH                0
        sulphates         0
        alcohol           0
        quality           0
        Id                0
        dtype: int64
```

Data Analysis and Visulaization

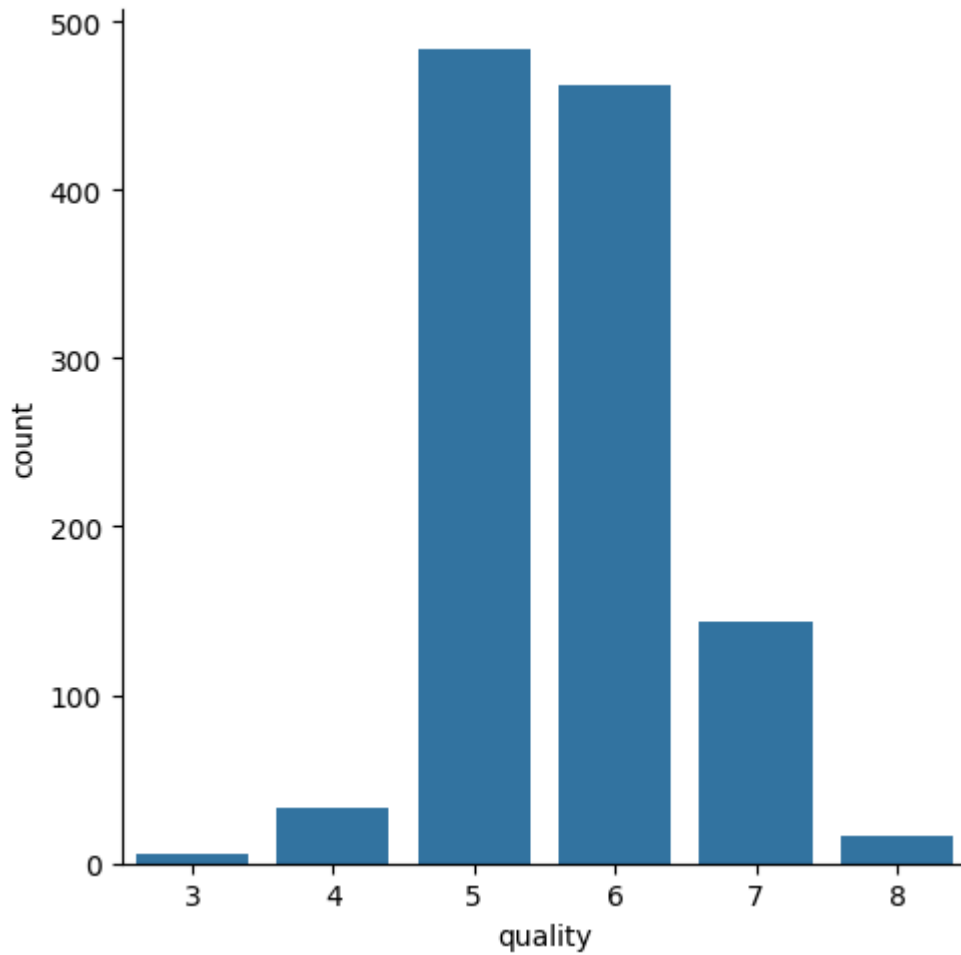
```
In [6]: #ststistical measures of the dataset
        wine_dataset.describe()
```

```
Out[6]:
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	
count	1143.000000	1143.000000	1143.000000	1143.000000	1143.000000	1143.000000	1
mean	8.311111	0.531339	0.268364	2.532152	0.086933	15.615486	
std	1.747595	0.179633	0.196686	1.355917	0.047267	10.250486	
min	4.600000	0.120000	0.000000	0.900000	0.012000	1.000000	
25%	7.100000	0.392500	0.090000	1.900000	0.070000	7.000000	
50%	7.900000	0.520000	0.250000	2.200000	0.079000	13.000000	
75%	9.100000	0.640000	0.420000	2.600000	0.090000	21.000000	
max	15.900000	1.580000	1.000000	15.500000	0.611000	68.000000	

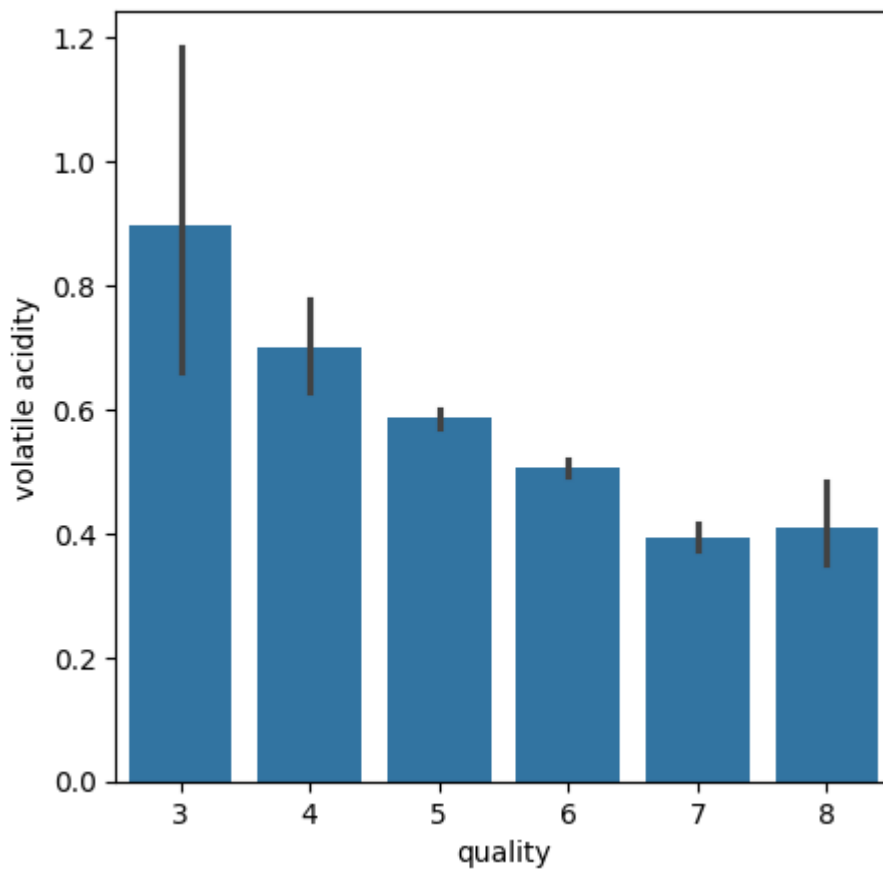
```
In [7]: #number of values for each quality
        sns.catplot(x='quality',data=wine_dataset,kind='count')
```

```
Out[7]: <seaborn.axisgrid.FacetGrid at 0x1dd3546fe60>
```



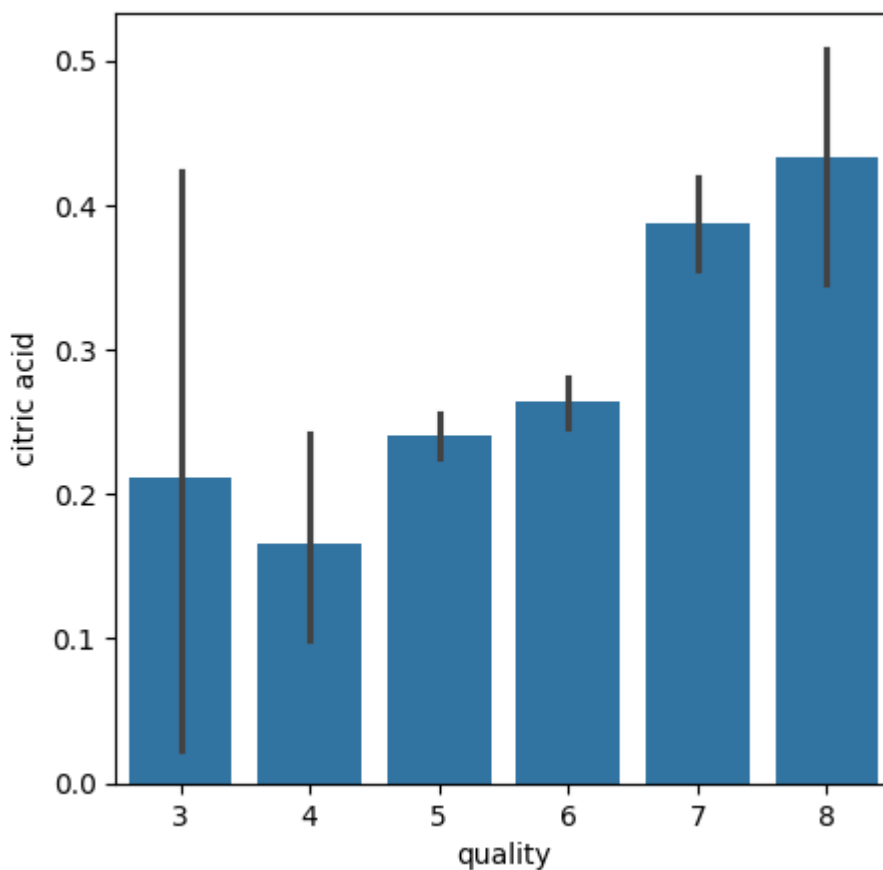
```
In [8]: #volatile acidity vs Quality
plot=plt.figure(figsize=(5,5))
sns.barplot(x='quality',y='volatile acidity',data=wine_dataset)
```

```
Out[8]: <Axes: xlabel='quality', ylabel='volatile acidity'>
```



```
In [9]: #citric acid vs Quality
plot=plt.figure(figsize=(5,5))
sns.barplot(x='quality',y='citric acid',data=wine_dataset)
```

```
Out[9]: <Axes: xlabel='quality', ylabel='citric acid'>
```



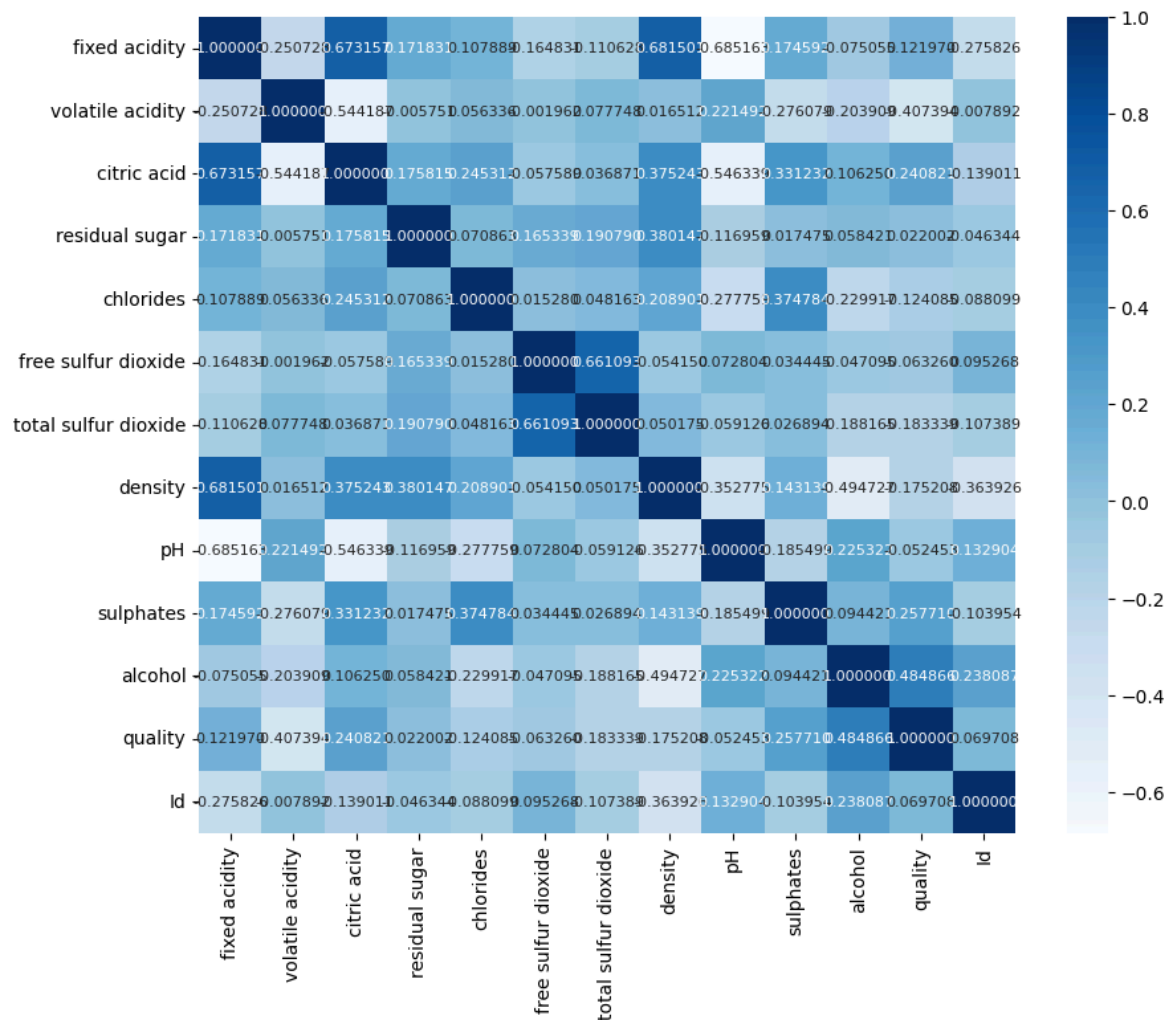
Correlation

1.Positive Correlation 2.Negative Correlation

```
In [10]: correlation=wine_dataset.corr()
```

```
In [11]: #constructing a heatmap to understand the correlation between the columns
plt.figure(figsize=(10,8))
sns.heatmap(correlation,cbar=True,square=True,fmt='1f',annot=True,annot_kws={'si
```

```
Out[11]: <Axes: >
```



Data Preprocessing

```
In [13]: #separate the data and label
x=wine_dataset.drop('quality',axis=1)
```

```
In [14]: print(x)
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	\
0	7.4	0.700	0.00	1.9	0.076	
1	7.8	0.880	0.00	2.6	0.098	
2	7.8	0.760	0.04	2.3	0.092	
3	11.2	0.280	0.56	1.9	0.075	
4	7.4	0.700	0.00	1.9	0.076	
...	...	...	...	...	...	
1138	6.3	0.510	0.13	2.3	0.076	
1139	6.8	0.620	0.08	1.9	0.068	
1140	6.2	0.600	0.08	2.0	0.090	
1141	5.9	0.550	0.10	2.2	0.062	
1142	5.9	0.645	0.12	2.0	0.075	

	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	\
0	11.0	34.0	0.99780	3.51	0.56	
1	25.0	67.0	0.99680	3.20	0.68	
2	15.0	54.0	0.99700	3.26	0.65	
3	17.0	60.0	0.99800	3.16	0.58	
4	11.0	34.0	0.99780	3.51	0.56	
...	...	...	...	...	...	
1138	29.0	40.0	0.99574	3.42	0.75	
1139	28.0	38.0	0.99651	3.42	0.82	
1140	32.0	44.0	0.99490	3.45	0.58	
1141	39.0	51.0	0.99512	3.52	0.76	
1142	32.0	44.0	0.99547	3.57	0.71	

	alcohol	Id
0	9.4	0
1	9.8	1
2	9.8	2
3	9.8	3
4	9.4	4
...	...	...
1138	11.0	1592
1139	9.5	1593
1140	10.5	1594
1141	11.2	1595
1142	10.2	1597

[1143 rows x 12 columns]

Label Binarization

```
In [15]: y=wine_dataset['quality'].apply(lambda y_value:1 if y_value>=7 else 0)
```

```
In [16]: print(y)
```

```
0      0
1      0
2      0
3      0
4      0
..
1138   0
1139   0
1140   0
1141   0
1142   0
Name: quality, Length: 1143, dtype: int64
```

Train & Test Split

```
In [18]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=3)
```

```
In [19]: print(y.shape,y_train.shape,y_test.shape)
```

```
(1143,) (914,) (229,)
```

Model Training:

Random Forest Classifier

```
In [20]: model=RandomForestClassifier()
```

```
In [22]: model.fit(x_train,y_train)
```

```
Out[22]:
```

▼ RandomForestClassifier ⓘ ?

- ▶ Parameters
- ▶ Fitted attributes

Model Evaluation

Accuracy Score

```
In [23]: #accuracy on test data
x_test_prediction=model.predict(x_test)
test_data_accuracy=accuracy_score(x_test_prediction,y_test)
print('Accuracy:',test_data_accuracy)
```

```
Accuracy: 0.9126637554585153
```

Building a Predictive System

```
In [29]: input_data=(7.8,0.43,0.32,2.8,0.08,29,58,0.9974,3.31,0.64,10.3,5)

#changing the input data to a numpy array
input_data_as_numpy_array=np.asarray(input_data)

#reshape the data as we are predicting the label for only one instance
input_data_resaped=input_data_as_numpy_array.reshape(1,-1)

prediction=model.predict(input_data_resaped)
print(prediction)

if(prediction[0]==1):
    print('Good Quality Wine')
else:
    print('Bad Quality Wine')
```

```
[0]
Bad Quality Wine
```

```
C:\Users\Admin\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn  
\utils\validation.py:2827: UserWarning: X does not have valid feature names, but  
RandomForestClassifier was fitted with feature names  
warnings.warn(  

```

In []: